

TALK IS CHEAP



Cut costs by using the Internet to place voice calls

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Did you know your business can cut down on phone bills using the Internet? Internet telephony or Voice over Internet Protocol (VoIP) is a means of transmitting voice data over the Internet. This technology offers a cheap alternative to your landline, and is ideal for placing long-distance conference calls.

We take a brief look at how VoIP works, the government regulations involved, and what you need to use it. We also walk you through the installation and use of a VoIP software called Skype.

Can You Hear Me Now?

VoIP transmits voice wrapped into data packets. A voice call made using VoIP is routed over a packet-switched network, whereas a traditional phone call takes place over a circuit-switch network. This is the essential difference between a VoIP call and a regular call over a land line every pro and each con of a VoIP service is a result of this. So what is a packet-switched network? When you make a regular phone call, a dedicated connection is made between the two parties involved. This connection is fixed at the two end-points, and has a fixed route over the phone line. The dedicated connection is maintained until the call ends. A packet-switched network, on the other hand, does not establish a fixed route between the two parties. Each packet can take a different route to its destination; packets can even arrive out of order at the destination. A route is taken on the basis of algorithms that balance various network factors such as latency and hops.

To better understand how a VoIP call works, let's take a look at two parties talking over their PCs. Say Arvind needs to make a call to Swati across the national branches of their business. Let's assume that both Arvind and Swati have Google Talk installed, and that they are listed in each other's contact lists. When Arvind clicks on Swati's username and initiates the call, the software first locates Swati on its network. If Swati answers the call, the software notes her destination address.

When Arvind talks, the voice data involved could be compressed using a proprietary algorithm to facilitate better transfer over the Internet. The compressed voice data is then made into an IP packet each IP packet is essentially a chunk of voice data wrapped around the destination address. The packet is then sent over the routers, switches, servers and cables that constitute the Internet: they hop between routers and switches, changing paths, perhaps taking a physically longer route to avoid network congestion, thus reducing latency, which you can think of as delay.

Each packet might also be tagged with data other than its destination address such as the maximum number of hops (the jumps it makes along the way) it can take, or how important latency is to that particular packet, and so on. It is important to note that each packet need not arrive at the same order at its destination. When packets do arrive at Swati's computer, Google Talk handles three things: the shuffling of the packets such that they get into the same order as they were sent from Arvind, the decompression of the packets to get to the raw voice data, and the transmission of the voice data to Swati's speakers or headset.

Several things happened differently here: no phone number was involved, voice was sent over the Internet rather than a dedicated phone line, and depending upon the quality of the Internet connection between Arvind and Swati, voice packets might have dropped, resulting in bad voice reception.

This last point falls under the purview of Quality of Service (QoS), and is perhaps the largest difference between a VoIP connection and a Plain Old Telephone Service (POTS) one. A POTS service is built upon a circuit-switching network, and has basic QoS guidelines to follow by law. While we might not appreciate the quality over our landlines, it could be worse; indeed, a VoIP connection might or might not ensure a good voice connection. However, some ISPs offer VoIP as part of their service bouquet. These ISPs might ensure a minimal QoS, in which case we recommend you take the service for a test ride before subscribing to an ISP's VoIP service.

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